Config and Lambda remediation

AWS Academy is the main lesson. This short deck connects it to our class work.

PRIMARY

AWS Academy

Cloud Security Foundations · Module 7

IN CLASS

Practice

Lab 7.1 · Config / Lambda (75 min)

COURSE WORK

Local task

Local 10 · C5R1 Part 2 / C5R2 Part 1

Local 10 closes 8 Mar · Use only your assigned Part

One path through the topic

STEP 1

Observe

AWS Config records the security-group configuration change.

STEP 2

Remediate

A custom rule invokes Lambda to restore the intended inbound rules.

STEP 3

Prove

Check the final security group and CloudWatch Logs, not only function success.

What proves the dangerous rule was removed?

What to do this week

BEFORE CLASS

Prepare

Open Module 7 and the remediation lab.

IN CLASS

Work

Complete Lab 7.1 and compare before-and-after rules.

AFTER CLASS

Record

Use slides 5–7 and complete Local 10 outside class by 8 Mar.

AWS work stays in AWS Academy · Local work stays in Classroom

Outside class / course closure

LAB EVIDENCE

Verify

Keep final security-group rules and relevant CloudWatch log events.

LOCAL 10

Submit

Use the local lab and assigned Part; turn in by 8 Mar at 23:59.

REFLECTION

Explain

State what failed, what changed, and how the final state was checked.

Local 09 late submissions also close 8 Mar at 23:59.

Detect, remediate, verify

CONFIG

Evaluate

A Config rule marks a resource compliant or noncompliant; that result does not itself change the resource.

LAMBDA

Enforce allowlist

A configured route invokes remediation. Keep ports 80 and 443; remove disallowed inbound rules such as port 22.

CHECK

Inspect final rules

A successful function run does not prove the rule set is safe. Compare before and after, then inspect logs.

Local 10: trace each rule through the allowlist code before calling it a fix.

Which rules survive the condition?

INPUT

80 · 443 · 22

The intended allowlist is {80, 443}; port 22 is open to everyone in the local starting state.

INVERTED TEST

Keep port NOT in list

The vulnerable function keeps 22 and removes 80/443. It can report success while leaving danger.

CORRECT TEST

Keep port IN list

The fixed function keeps 80/443 and removes 22. Compare actual rules after the run.

Trace the worksheet's supplied code by hand before answering Part 2b.

A successful run is only one signal

DETECTION

Config evaluation

A NON_COMPLIANT result reports a state; a separately configured path must invoke remediation.

SAFETY

Port 22 removed?

Inspect the final security-group rules; a 200 response or Lambda log alone cannot prove this.

SERVICE

80/443 still work?

Check the intended access too. Removing safe rules while retaining SSH is a failed security outcome.

Local Flask targets model rule logic; AWS Config inventory and timing are not simulated.